

DELIVERY OPERATIONS & PROFITABILITY ANALYSIS

*Root-Cause Analysis of Late Deliveries and Their Financial Impact, Jan 2015 –
Jan 2018*

Global E-Commerce Order Fulfillment | 1,72,765 Orders
Operations & Supply Chain Analytics Report
Prepared: June 2026

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1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 1,72,765 orders spanning January 2015 through January 2018, with the objective of identifying the root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for improvement.

The headline findings are striking: 54.71% of all orders are delivered late, costing the business an estimated \$2.1M in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached \$7.5M, the persistent late-delivery problem represents both a significant financial drain and a major customer-experience risk.

Key Findings

Metric	Value
Total Orders Analyzed	1,72,765
Late Deliveries	94,523
Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Profit (Profitable Orders)	\$7.5M
Profit at Risk (Delayed Orders)	\$2.1M
90th Percentile Delay	3 days
Average Order Profit	\$22.03

2. Business Context and Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel, serving customers across multiple international regions.

Core Problem

Actual shipping times frequently deviate from scheduled delivery windows, eroding customer trust, reducing order profitability, and undermining the company's ability to make reliable delivery commitments at the point of purchase.

Analytical Objectives

- Understand the current state of delivery performance across all key dimensions — region, shipping mode, time, and customer segment.
- Quantify the financial impact of delays on order profitability.
- Identify the primary operational bottlenecks driving late deliveries.
- Build a predictive model to flag high-risk orders before they are shipped (advanced).
- Deliver actionable recommendations to improve on-time delivery and profitability.

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline for the fulfillment operation.

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Interpretation

- **Late delivery rate of 54.71%:** more than half of all orders arrive late. This is not an edge-case problem — it represents the default experience for the majority of customers.
- **\$2.1M profit at risk:** orders that experienced delays collectively generated \$2.1M in profit that remains under constant pressure from further operational deterioration.
- **90th percentile delay of 3 days:** even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than caused by rare outliers.

4. Profitability Analysis

4.1 Profit Distribution

Order-level profitability was classified into three tiers based on profit per order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag on the business — one that is disproportionately concentrated among delayed shipments.

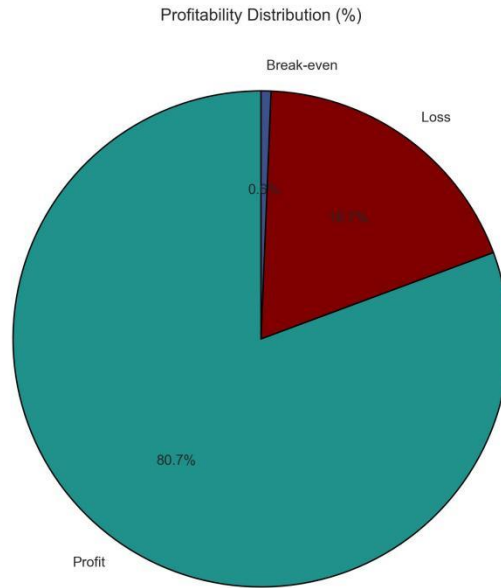


Figure 1: Profitability Distribution — 80.7% of orders are profitable, 18.7% result in a loss, and 0.6% break even.

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly one day late — the single largest cohort. Combined, orders delayed by one to four days account for 54.7% of total order volume, directly mapping to the overall late-delivery rate.



Figure 2: Delay Distribution (left) and Total / Mean Profit by Delay Days (right).

Observation: mean profit per order remains stable at approximately \$20–\$23 across all delay levels. The profitability problem is therefore driven by the volume of delayed orders, not by the economics of individual orders — making systemic throughput improvement the highest-leverage intervention available to the business.

5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfillment process breaks down most severely.

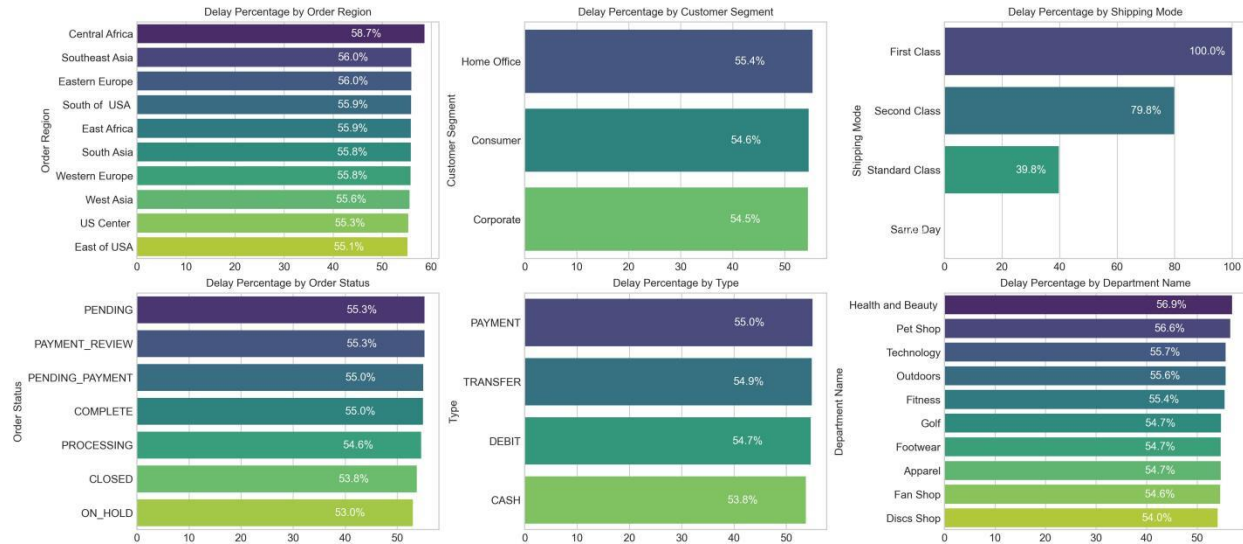


Figure 3: Delay Percentage by Order Region, Customer Segment, Shipping Mode, Order Status, Type, and Department.

Key Observations

- **Shipping mode is the #1 lever:** First Class has a 100% delay rate, Second Class 79.8%, Standard Class 39.8%, and Same Day 0%. Shipping-mode assignment logic is the single most impactful variable to fix.
- **Regional spread is narrow (55–59%):** Central Africa leads at 58.7%, but all regions cluster within a 55–59% band. This rules out a localized failure and confirms a company-wide systemic issue.
- **Customer segments are nearly identical (54.5–55.4%):** no segment receives preferential service — Consumer, Corporate, and Home Office customers all experience the same broken delivery promise.
- **Health & Beauty (56.9%) and Pet Shop (56.6%) lead by department:** these departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these product categories.

6. Root Cause Analysis

Central Africa was selected for a deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top ten driver factors by delay percentage within this region.

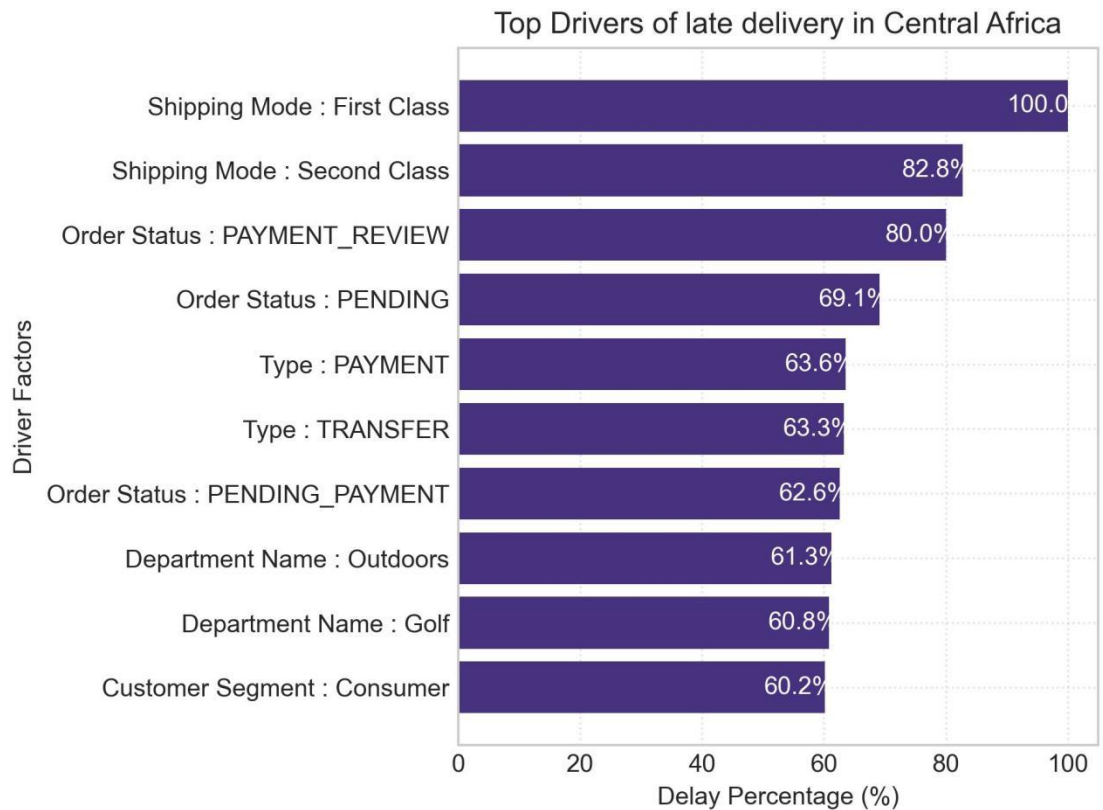


Figure 4: Top 10 Drivers of Late Delivery in Central Africa.

The pattern in Central Africa mirrors — and amplifies — the company-wide findings: First Class shipping reaches a 100% delay rate, Second Class 82.8%, and orders held in PAYMENT_REVIEW status are delayed 80.0% of the time. Outdoors (61.3%) and Golf (60.8%) departments also rank among the top drivers, indicating a region-specific inventory and carrier issue layered on top of the systemic shipping-mode and payment-processing problems.

7. Time-Based Analysis

Three temporal dimensions were analyzed: month of year, day of week, and hour of day. While delay rates are relatively stable across all dimensions (53–57%), specific peaks highlight opportunities for targeted capacity planning.

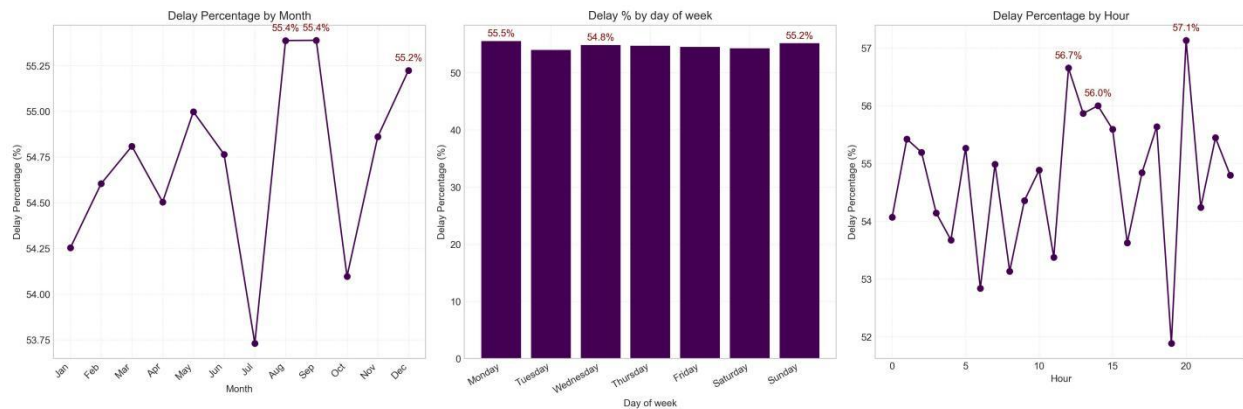


Figure 5: Delay Percentage by Month, Day of Week, and Hour of Day.

Key Observations

- **Peak delay months — August & September (55.4%) and December (55.2%):** these months reflect mid-year promotions and Q4 holiday surges overwhelming fulfillment capacity. July is a notable low point (53.7%), confirming that seasonal variation is predictable and plannable.
- **Day of week — variance is minimal:** Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than one percentage point. Day of week is not a meaningful lever for operational intervention.
- **Intra-day peaks — hour 21 (57.1%), hour 11 (56.7%), hour 12 (56.0%):** late-evening orders (hour 21) show the highest delay rate, likely reflecting end-of-day processing cutoff windows. Midday peaks suggest bottlenecks during high-volume order hours.

8. Strategic Recommendations

Based on findings across all analytical dimensions, the following prioritized actions are recommended.

#	Priority	Action	Rationale
1	Critical	Audit First & Second Class shipping capacity	First Class shows a 100% delay rate and Second Class 79.8% — both operate in direct contradiction to their service promise. Commission an immediate carrier SLA audit; consider suspending First Class and repricing both modes to reflect actual performance.
2	High	Resolve payment-processing bottlenecks	Orders in PAYMENT_REVIEW (55.3% overall, 80.0% in Central Africa) and PENDING_PAYMENT (55.0%) show elevated delay rates. Introduce automated escalation for payment reviews stalled beyond 2 hours.
3	Medium	Build seasonal surge-capacity plans	August, September (55.4%) and December (55.2%) are peak delay months. Establish pre-season capacity plans, including temporary logistics partnerships, inventory pre-positioning, and adjusted delivery-promise windows.
4	Medium	Default eligible orders to Standard Class	Standard Class delivers a markedly lower 39.8% delay rate. Re-evaluate shipping-mode assignment logic to default eligible orders to Standard Class, particularly in regions and departments with poor premium-mode performance.
5	Medium	Audit Outdoors and Golf fulfillment in Central Africa	Outdoors (61.3%) and Golf (60.8%) show significantly elevated delay rates in Central Africa. Conduct a warehouse-level audit of stock availability, pick-and-pack times, and carrier assignment for these categories.
6	Low	Reduce the loss-making order share (18.7%)	These orders warrant a separate pricing and discount review. Cross-reference with delay analysis to determine whether discounted orders are disproportionately late, compounding the margin problem.

9. Conclusion

This analysis surfaces a clear and urgent picture: a global e-commerce operation in which the majority of orders (54.71%) fail to meet their promised delivery windows, placing \$2.1M in profit at risk and undermining customer trust at scale.

The root causes are identifiable and addressable. Shipping-mode misconfiguration — First Class at a 100% delay rate and Second Class at 79.8% — is the dominant operational failure. Payment-processing friction creates a secondary bottleneck, seasonal volume spikes are not adequately planned for, and geographic disparities, while present, are secondary to the systemic, company-wide pattern.

Addressing shipping-mode assignment and payment-review escalation alone has the potential to materially reduce the 54.71% late-delivery rate and protect the \$2.1M of profit currently at risk, while seasonal capacity planning and targeted departmental audits offer further incremental gains.